

Industrial Internship Report on "Crop Yield Prediction Using Machine Learning"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Crop Yield Prediction Using Machine Learning". The aim of this project was to predict agricultural crop yield based on various environmental and farming factors such as rainfall, temperature, pesticide usage, and cultivated area. The project uses machine learning techniques to analyze historical agricultural data and generate accurate yield predictions.

The developed model helps farmers and agricultural organizations make informed decisions regarding crop planning and resource management. By providing accurate yield predictions, the system can contribute to improved agricultural productivity and better utilization of available resources.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

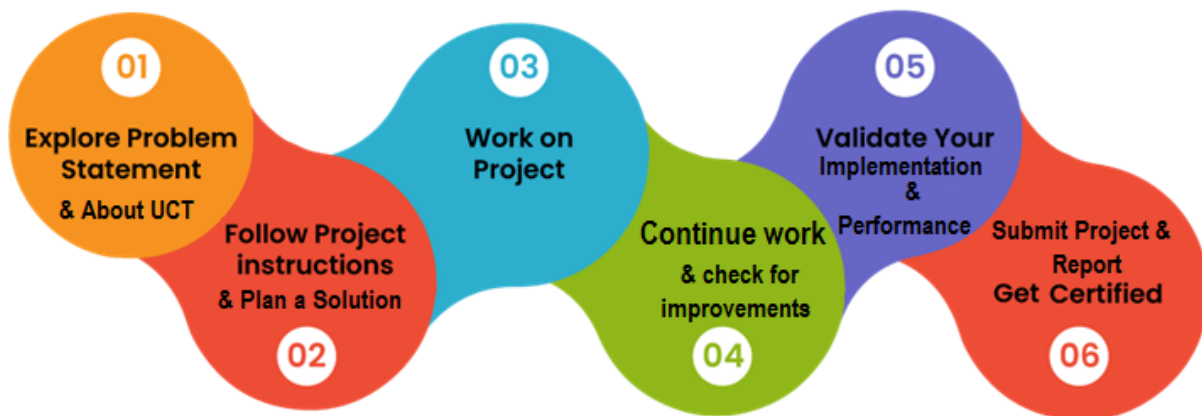
This report presents the work completed during my six-week Industrial Internship conducted by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd.

The internship provided me with valuable industry exposure and helped bridge the gap between academic learning and practical implementation. During this internship, I worked on the project "Crop Yield Prediction Using Machine Learning," which aims to predict crop yield using agricultural and environmental data.

The program was structured into different phases, including understanding the problem statement, data collection, data preprocessing, exploratory data analysis, machine learning model development, testing, and project submission.

This opportunity allowed me to enhance my technical skills in Python, Data Analysis, Machine Learning, and Problem Solving while gaining experience in handling real-world datasets.

How Program was planned



I would like to express my sincere gratitude to Upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd., and all the mentors and trainers for their continuous guidance and support throughout the internship.

My message to juniors and peers is to actively participate in internships and projects, as they provide practical knowledge, improve technical skills, and prepare students for future career opportunities.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform ()

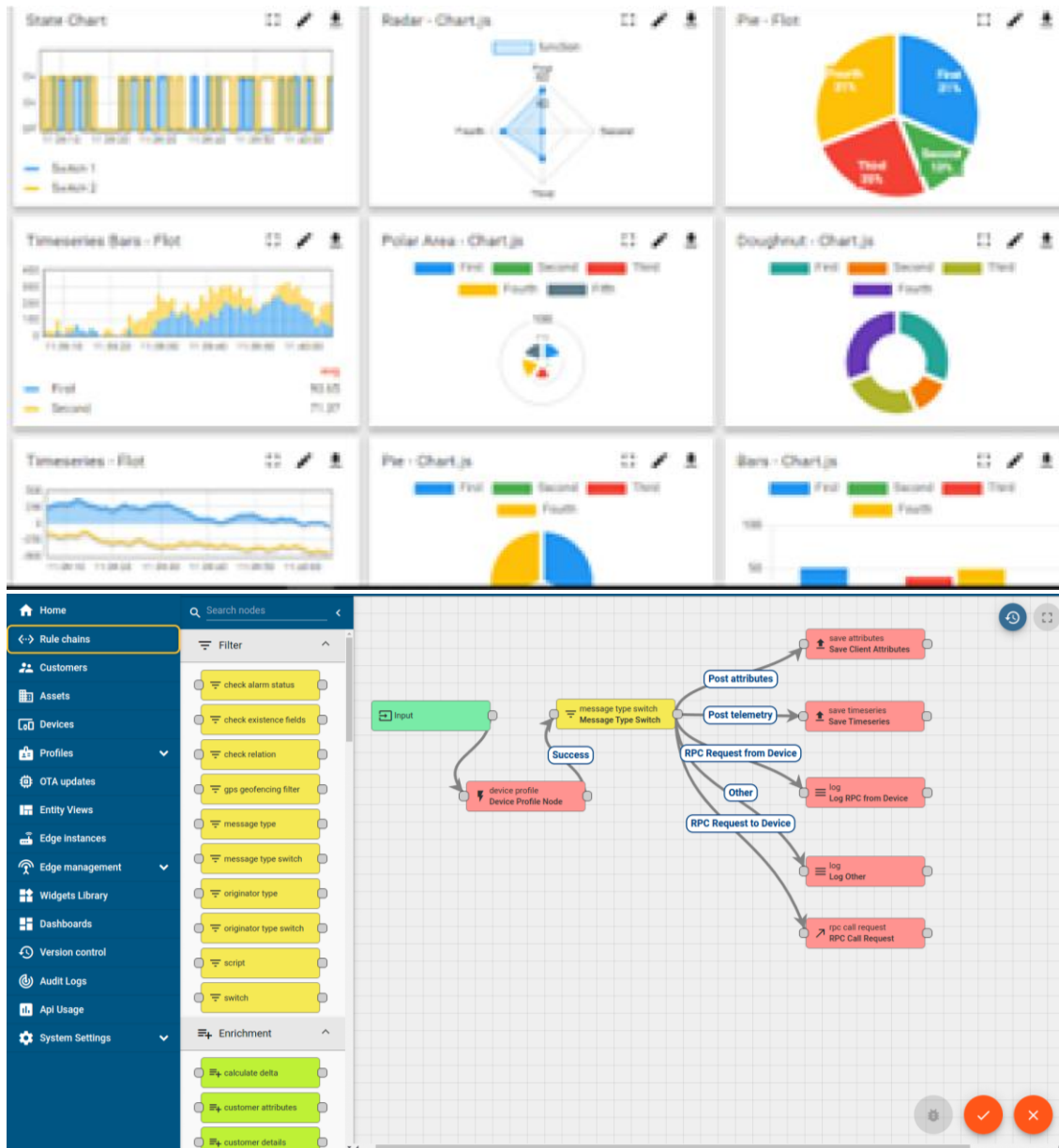
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Prod	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



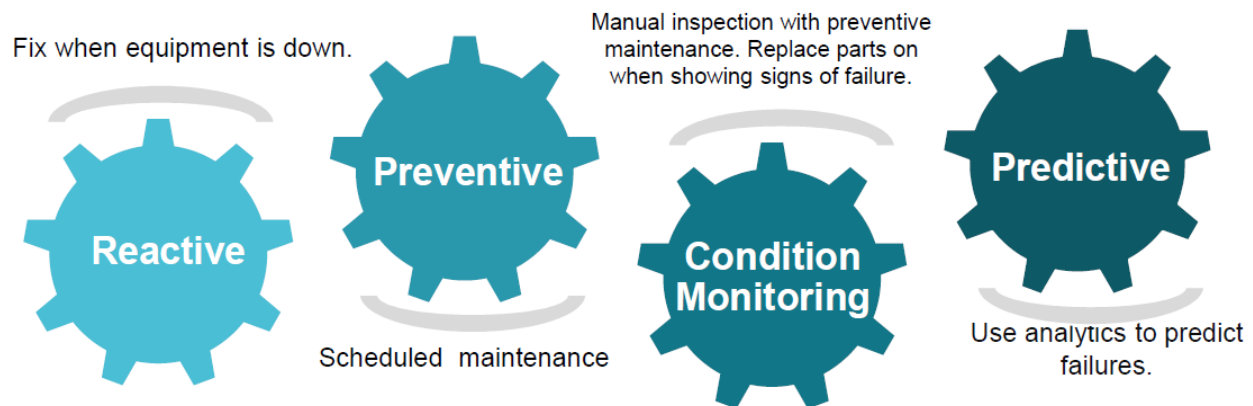


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

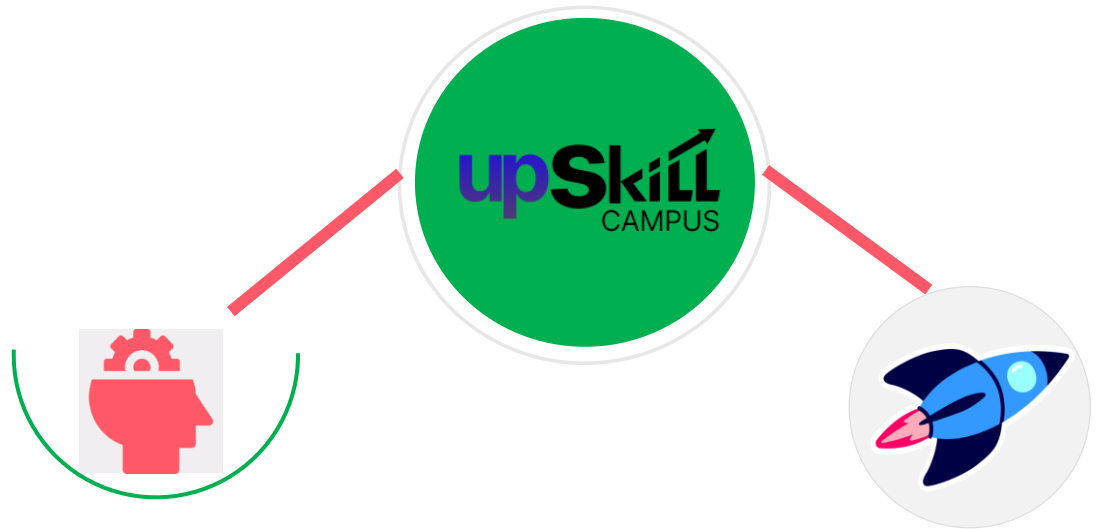
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

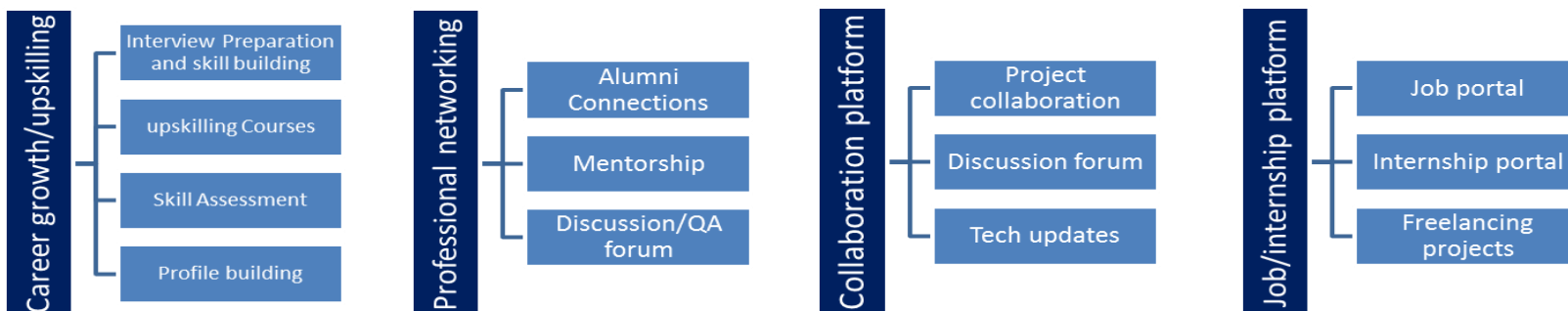
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Crop Yield Prediction Dataset – Kaggle
- [2] Scikit-Learn Documentation – Machine Learning Library for Python
- [3] Pandas and NumPy Official Documentation

2.6 Glossary

Terms	Acronym
Machine Learning	ML

Artificial Intelligence	AI
Exploratory Data Analysis	EDA
Random Forest	RF
Mean Absolute Error	MAE
Root Mean Square Error	MAE
Coefficient of Determination	R2
Dataset	DS

3 Problem Statement

Agriculture is one of the most important sectors of the economy and plays a vital role in ensuring food security and economic growth. Farmers often face challenges in estimating crop production due to various factors such as changing weather conditions, rainfall patterns, temperature variations, soil quality, and the use of fertilizers and pesticides. Inaccurate yield estimation can lead to poor planning, inefficient resource utilization, and financial losses.

Traditional methods of predicting crop yield are often based on historical observations and manual analysis, which may not provide accurate results under changing environmental conditions. With the increasing availability of agricultural data, machine learning techniques can be used to analyze large datasets and identify patterns that influence crop production.

The objective of this project is to develop a Crop Yield Prediction System using Machine Learning. The system utilizes historical agricultural and environmental data, including factors such as rainfall, temperature, cultivated area, and pesticide usage, to predict crop yield accurately. The project involves data preprocessing, exploratory data analysis, feature selection, model training, and performance evaluation.

By leveraging machine learning algorithms, the proposed solution aims to improve prediction accuracy and support farmers, agricultural organizations, and policymakers in making informed decisions. Accurate crop yield prediction can help optimize resource allocation, improve agricultural productivity, reduce risks, and contribute to sustainable farming practices.

The successful implementation of this project demonstrates how data-driven technologies can be applied to solve real-world agricultural challenges and assist in improving decision-making within the agricultural sector.

4 Existing and Proposed solution

Existing Solutions

Several traditional and modern approaches are used for crop yield estimation. Farmers often rely on historical records, manual calculations, and expert opinions to estimate crop production. Some agricultural organizations use statistical methods and basic forecasting techniques based on past trends. Although these methods provide approximate estimates, they are often unable to handle large datasets and changing environmental conditions effectively.

Limitations of Existing Solutions

- Depend heavily on manual analysis and human expertise.
- Accuracy is affected by unpredictable weather conditions.
- Difficult to process large amounts of agricultural data.
- Time-consuming and less scalable.

Proposed Solution

The proposed solution is a Machine Learning-based Crop Yield Prediction System. The system uses historical agricultural and environmental data such as rainfall, temperature, pesticide usage, and cultivated area to predict crop yield. Data preprocessing, feature analysis, and machine learning algorithms are applied to train a predictive model capable of generating accurate yield estimates.

Value Addition

- Provides faster and more accurate crop yield predictions.
- Reduces dependency on manual calculations.
- Helps farmers make informed decisions regarding crop planning.

4.1 Code submission (Github link)

<https://github.com/Magdarel/upskillcampus.git>

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The proposed Crop Yield Prediction System follows a structured machine learning workflow to predict crop yield using historical agricultural and environmental data. The system begins with data collection from agricultural datasets containing information such as rainfall, temperature, pesticide usage, crop type, and cultivated area. The collected data is then cleaned and preprocessed to remove missing values and inconsistencies.

After preprocessing, Exploratory Data Analysis (EDA) is performed to understand patterns and relationships among different features. Relevant features are selected and used for model training. The dataset is divided into training and testing sets, and machine learning algorithms are applied to develop the prediction model. The trained model is then evaluated using performance metrics to measure prediction accuracy. Finally, the model generates crop yield predictions that can assist farmers and agricultural organizations in decision-making and resource planning.

5.1 High Level Diagram (if applicable)

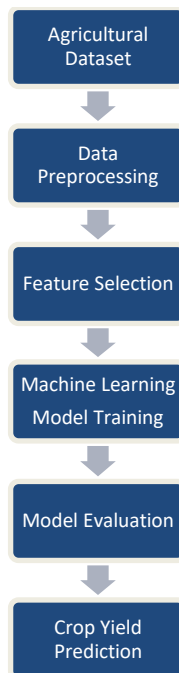


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

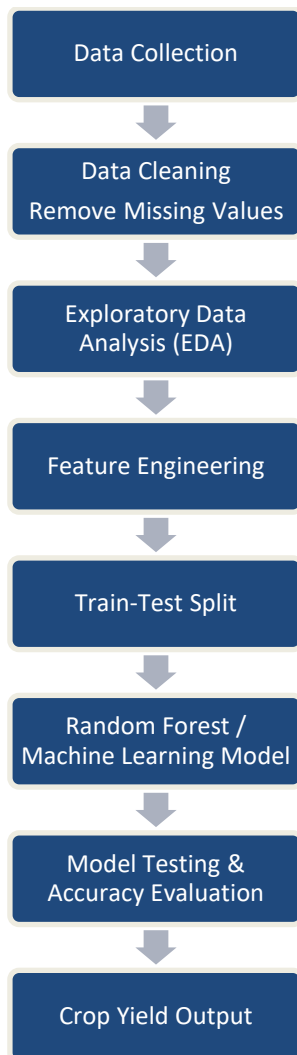


Figure 2: LOW LEVEL DIAGRAM OF CROP YIELD PREDICTION SYSTEM

5.3 Interfaces (if applicable)

The Crop Yield Prediction System was developed and executed using Jupyter Notebook and Python. The project interface consists of different stages, including dataset loading, data preprocessing, exploratory data analysis, machine learning model training, and prediction generation.

The user interacts with the system by providing agricultural and environmental data such as rainfall, temperature, pesticide usage, and cultivated area. The system processes the input data, applies the trained machine learning model, and generates the predicted crop yield as output.

Various visualization techniques such as graphs, charts, and statistical plots were used during data analysis to understand patterns in the dataset and evaluate model performance. The final output is displayed within the Jupyter Notebook environment, providing predicted crop yield values and performance metrics.

6 Performance Test

The performance of the Crop Yield Prediction System was evaluated using machine learning evaluation techniques. The model was tested on unseen data to determine its prediction accuracy and reliability. Various constraints such as data quality, prediction accuracy, processing time, and model efficiency were considered during the testing phase.

One of the major challenges was handling missing and inconsistent data in the dataset. Data preprocessing techniques were applied to improve data quality and ensure accurate model training. The model was also optimized to reduce prediction errors and improve overall performance.

The developed system successfully generated crop yield predictions based on agricultural and environmental factors. The test results demonstrated that machine learning can effectively analyze historical data and provide reliable yield estimates. The model achieved satisfactory performance and can assist farmers and agricultural organizations in making informed decisions regarding crop planning and resource management.

Constraints

- Missing values in dataset.
- Limited dataset size.
- Weather conditions can change unpredictably.
- Prediction accuracy depends on data quality.
- Different crops may have different yield patterns.

6.1 Test Plan/ Test Cases

The objective of testing was to verify the accuracy and reliability of the Crop Yield Prediction System. Various test cases were designed to ensure proper functioning of the machine learning model and data processing pipeline.

Test Case 1: Verify successful loading of the agricultural dataset.

Test Case 2: Check preprocessing of data, including handling of missing values and removal of inconsistencies.

Test Case 3: Validate feature selection and preparation of input variables for model training.

Test Case 4: Verify successful training of the regression model using the training dataset.

Test Case 5: Test the model using unseen test data and generate crop yield predictions.

Test Case 6: Compare predicted crop yield values with actual values to evaluate model performance.

Test Case 7: Verify that the system produces prediction results without errors and within acceptable processing time.

The successful execution of these test cases confirmed that the developed system performs crop yield prediction accurately and efficiently.

6.2 Test Procedure

The Crop Yield Prediction System was tested using a systematic approach to evaluate the performance of the regression model.

1. The agricultural dataset was collected and loaded into the Jupyter Notebook environment.
2. Data preprocessing was performed to handle missing values, remove inconsistencies, and prepare the dataset for analysis.
3. Exploratory Data Analysis (EDA) was conducted to understand the relationships between different features and crop yield.
4. Relevant features such as rainfall, temperature, pesticide usage, and cultivated area were selected for model training.
5. The dataset was divided into training and testing datasets using an appropriate train-test split ratio.
6. A regression-based machine learning model was trained using the training dataset.
7. The trained model was applied to the testing dataset to generate crop yield predictions.
8. Predicted values were compared with actual crop yield values to evaluate model performance.
9. Performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) were analyzed to measure prediction accuracy.
10. The final results were reviewed to determine the effectiveness and reliability of the developed prediction system.

The successful completion of the above procedure confirmed that the model was capable of generating crop yield predictions based on historical agricultural data.

6.3 Performance Outcome

The Crop Yield Prediction System was successfully developed and tested using a supervised machine learning regression model. The trained model was integrated with a Flask-based web application, allowing users to enter agricultural parameters such as Year, Average Rainfall, Pesticide Usage, Average Temperature, Area, and Crop Type.

During testing, the system successfully processed user inputs and generated crop yield predictions in real time. The web interface displayed the predicted yield value accurately after the user clicked the Predict button. The model demonstrated its ability to analyze agricultural data and estimate crop production based on historical patterns.

The successful implementation of the prediction system confirms that machine learning techniques can be effectively applied to agricultural forecasting. The developed solution provides a user-friendly platform for crop yield estimation and can assist farmers, researchers, and agricultural organizations in planning and decision-making. Future enhancements may include incorporating real-time weather data, soil parameters, and advanced machine learning algorithms to further improve prediction accuracy.

Result: The system successfully generated crop yield predictions through the web interface and produced reliable output based on the input parameters provided by the user.

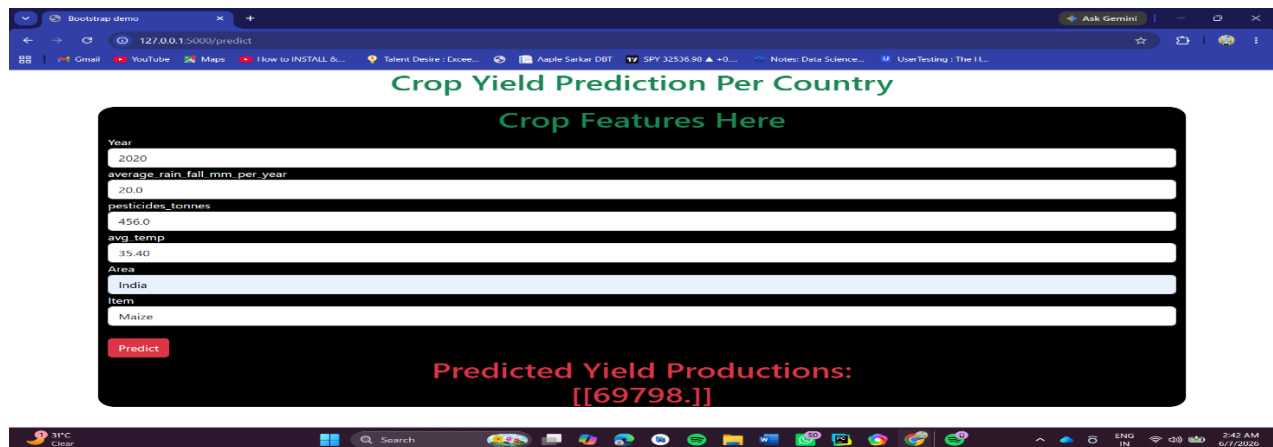


Figure 3: FINAL OUTPUT OF CROP YIELD PREDICTION SYSTEM

7 My learnings

During this internship, I gained practical knowledge of Machine Learning and its real-world applications in agriculture. I learned how to collect, clean, and preprocess datasets, perform Exploratory Data Analysis (EDA), and build predictive models using supervised learning techniques. Through the Crop Yield Prediction project, I understood the complete machine learning workflow, including feature selection, model training, testing, and performance evaluation.

I also improved my programming skills in Python and learned to use libraries such as Pandas, NumPy, Matplotlib, Scikit-Learn, and Flask. Developing a web-based interface helped me understand how machine learning models can be integrated into applications to provide real-time predictions for users.

This internship enhanced my problem-solving, analytical thinking, and project development skills. It provided hands-on experience in working with real datasets and implementing machine learning solutions for practical problems. The knowledge and experience gained during this internship will help me in my future career in Data Science, Machine Learning, and Software Development.

Key Learnings

- Machine Learning fundamentals and supervised learning.
- Data cleaning and preprocessing techniques.
- Exploratory Data Analysis (EDA).
- Regression model development and evaluation.
- Python programming and data science libraries.
- Flask web application development.
- Project documentation and GitHub usage.
- Real-world application of AI and Machine Learning.

8 Future work scope

The Crop Yield Prediction System developed during this internship successfully predicts crop yield using machine learning techniques. However, there is significant scope for further enhancement and improvement.

In the future, the system can be integrated with real-time weather data to improve prediction accuracy and provide more up-to-date results. Additional parameters such as soil quality, humidity, fertilizer usage, and groundwater availability can also be included to make the predictions more comprehensive

The project can be extended to support multiple machine learning and deep learning algorithms for better performance comparison and improved accuracy. A mobile application can also be developed to make the system easily accessible to farmers and agricultural experts.

Furthermore, the system can be integrated with cloud platforms and IoT devices to collect real-time field data automatically. This would enable continuous monitoring and more accurate crop yield forecasting. Future versions may also include crop recommendation and disease prediction features, making the platform a complete smart agriculture solution.

Overall, the project provides a strong foundation for the application of Artificial Intelligence and Machine Learning in agriculture and offers several opportunities for future development and research.

- Integration with real-time weather APIs.
- Inclusion of soil quality and humidity data.
- Development of a mobile application.
- Deployment on cloud platforms.
- Use of advanced machine learning and deep learning models.
- Crop recommendation system.
- Crop disease detection and prediction.
- IoT-based smart farming integration.

